DAY_7

- **VLAN Concepts Theory: VLANs: Segmentation, benefits, and use cases. Practical : Create VLANs on a switch and assign ports. Task: Document VLAN configurations.**

1. On a Cisco Packet Tracer switch, create two VLANs named 'Engineering' (VLAN 10) and 'Marketing' (VLAN 20) using the CLI commands.

ANS :

- To understand VLAN segmentation, create VLANs on a Cisco switch, assign switch ports to different VLANs, document the configuration, and understand the benefits of VLANs in a college campus network.

  - ### Create Two VLANs :

VLAN ID	VLAN Name
10	Engineering
20	Marketing

- ### Cisco Packet Tracer CLI Commands

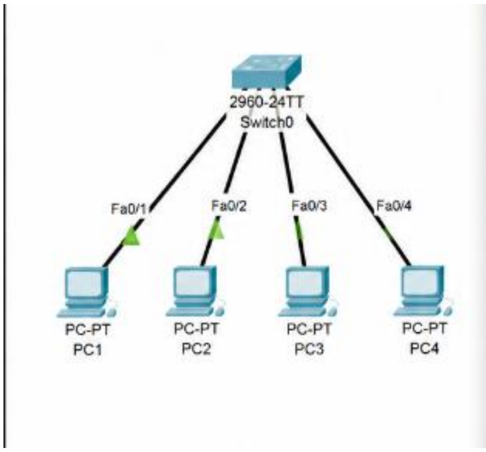
  Enter privileged mode:
  Switch> enable
  Switch# configure terminal

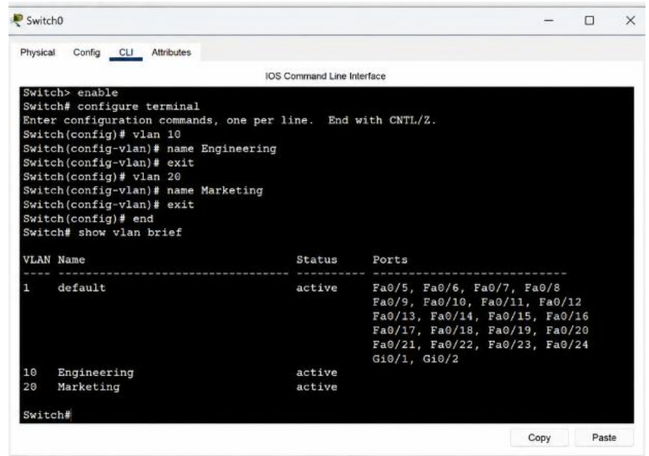
- ### Create VLAN 10 — Engineering

  Switch(config)# vlan 10
  Switch(config-vlan)# name Engineering
  Switch(config-vlan)# exit

- ### Create VLAN 20 — Marketing

  Switch(config)# vlan 20
  Switch(config-vlan)# name Marketing
  Switch(config-vlan)# exit





2. Assign ports FastEthernet0/1 and FastEthernet0/2 to VLAN 10, and ports FastEthernet0/3 and FastEthernet0/4 to VLAN 20 on your switch.

ANS :

- **Assign Fa0/1 and Fa0/2 to VLAN 10**

  Switch(config)# interface range fastEthernet 0/1 - 2
  Switch(config-if-range)# switchport mode access
  Switch(config-if-range)# switchport access vlan 10
  Switch(config-if-range)# exit

- **Assign Fa0/3 and Fa0/4 to VLAN 20**

  Switch(config)# interface range fastEthernet 0/3 - 4
  Switch(config-if-range)# switchport mode access
  Switch(config-if-range)# switchport access vlan 20
  Switch(config-if-range)# exit
  Save the configuration:
  Switch(config)# end
  Switch# write memory
  or:
  Switch# copy running-config startup-config

3. Write a short document (in a text file) listing the VLAN IDs, names, and the ports assigned to each VLAN from your configuration.

ANS :

- You can put the following content into a text file such as **VLAN_Configuration.txt**:

```
VLAN CONFIGURATION DOCUMENT
===========================

Switch: Cisco Packet Tracer Switch

VLAN 10
-------
VLAN Name : Engineering
Ports     : FastEthernet0/1, FastEthernet0/2

VLAN 20
-------
VLAN Name : Marketing
Ports     : FastEthernet0/3, FastEthernet0/4


Summary
-------
VLAN 10 - Engineering - Fa0/1, Fa0/2
VLAN 20 - Marketing   - Fa0/3, Fa0/4
```

- **Configuration Summary**

VLAN ID	VLAN Name	Assigned Ports
10	Engineering	Fa0/1, Fa0/2
20	Marketing	Fa0/3, Fa0/4

4. Explain, in your own words, two benefits of using VLANs in a college campus network that has separate labs for coding and networking.

ANS :

- **Benefit 1: Network Segmentation**

  - VLANs can separate different departments or labs into different logical networks.
    For example:
    Coding Lab → VLAN 10
    Networking Lab → VLAN 20

  - Devices in the Coding Lab remain logically separated from devices in the Networking Lab. This reduces unnecessary broadcast traffic and makes the network easier to manage.

- **Benefit 2: Improved Security**

  - VLANs can help restrict communication between different groups.
    For example, computers in the Coding Lab do not automatically need direct access to all computers in the Networking Lab. Communication between VLANs can be controlled using a **router or Layer 3 switch** and appropriate access-control rules.

  - This helps protect important systems and reduces the possibility of unauthorized access.

Cisco Switch

VLAN 10	VLAN 20
Engineering	Marketing
Fa0/1	Fa0/3
Fa0/2	Fa0/4